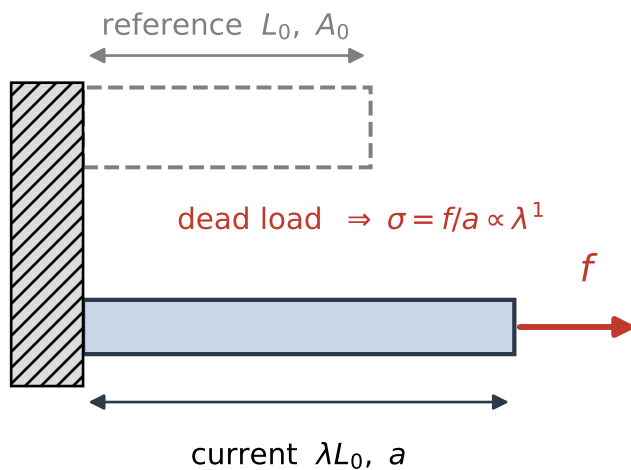
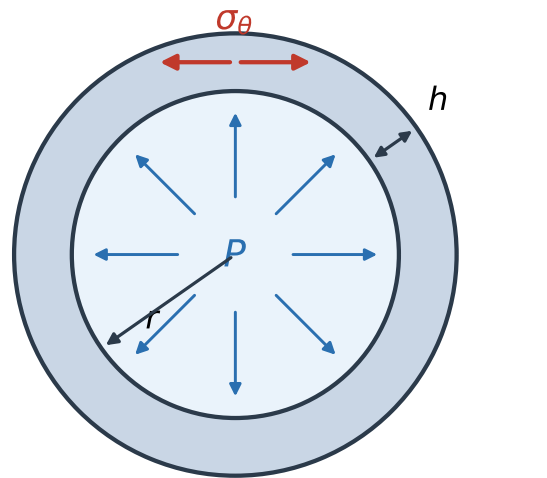


Shared setting:  $R = 1.0 \text{ mm}$ ,  $H = 130 \mu\text{m}$ ,  $P_h = 13.3 \text{ kPa}$ ,  $\bar{\sigma}_h = 102 \text{ kPa}$

(a) 1-D tissue bar



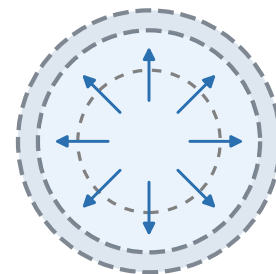
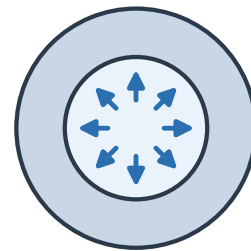
(b) thin-walled artery (cross-section)



Laplace:  $\sigma_\theta = Pr/h \propto \lambda^2 \odot z, \lambda_z$

(c) the two insults

hypertension:  $P \uparrow \Rightarrow$  wall thickens



aneurysm: elastin lost  $\Rightarrow$  dilation